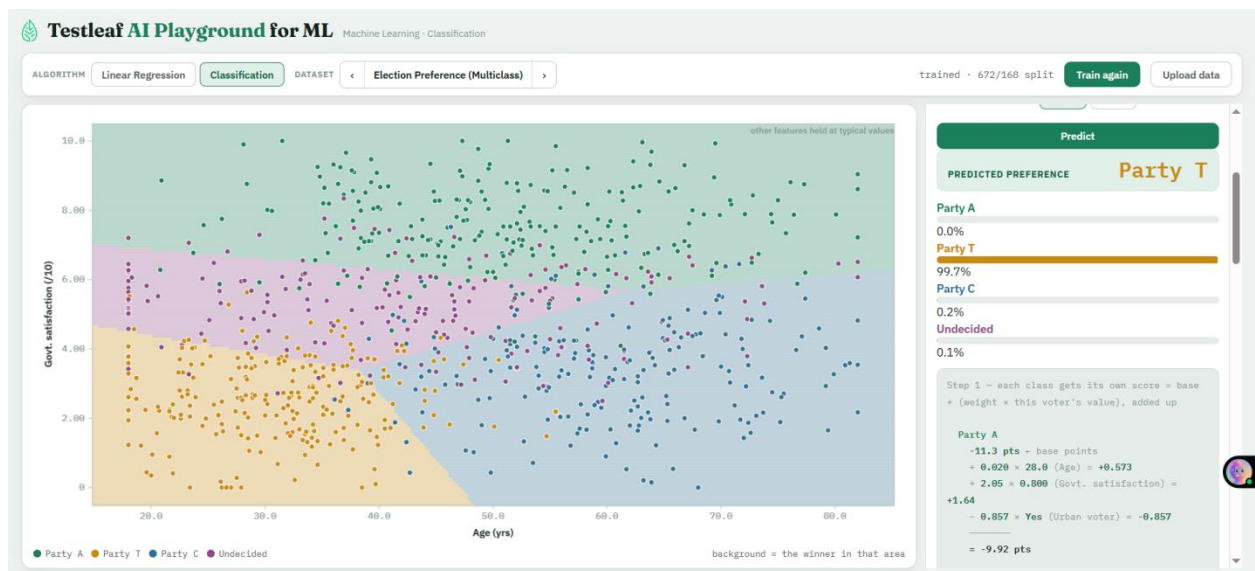
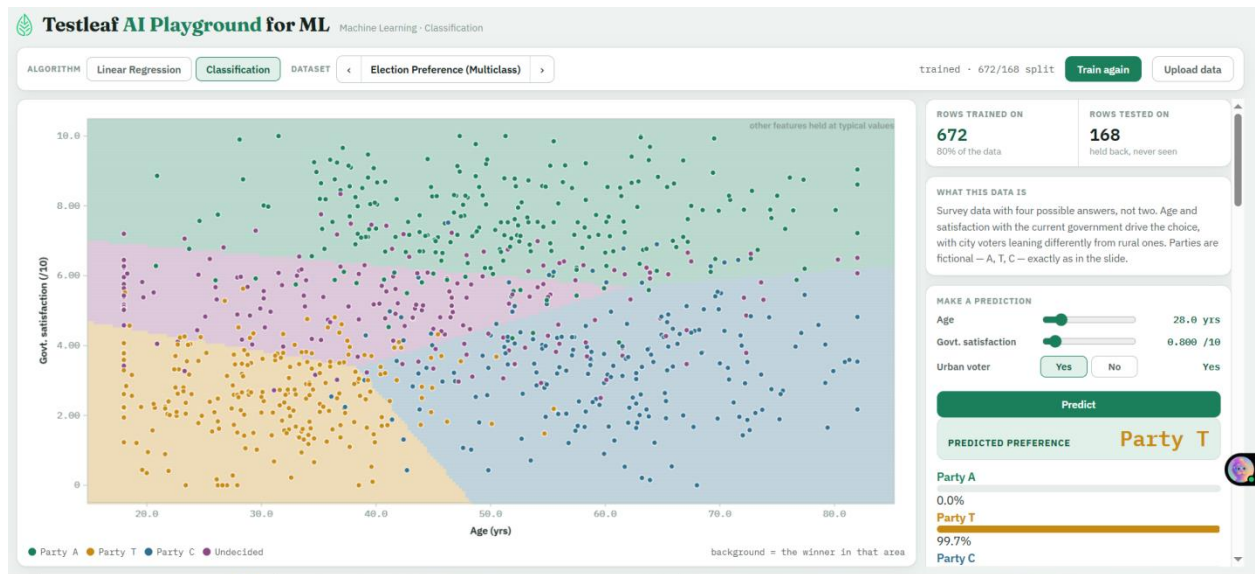


1) Classification Algorithm

a) Age <30 and Govt satisfaction less ,urban



ALGORITHM Linear Regression Classification DATASET Election Preference (Multiclass)

trained · 672/168 split Train again Upload data



Party T

11.5 pts = base points
 -0.130×28.0 (Age) = -3.65
 -1.49×0.800 (Govt. satisfaction) = -1.19
 $+1.30 \times \text{Yes (Urban voter)}$ = $+1.30$
 = **7.92 pts**

Party C

0.764 pts = base points
 $+0.097 \times 28.0$ (Age) = $+2.71$
 -0.902×0.800 (Govt. satisfaction) = -0.721
 $-1.14 \times \text{Yes (Urban voter)}$ = -1.14
 = **1.62 pts**

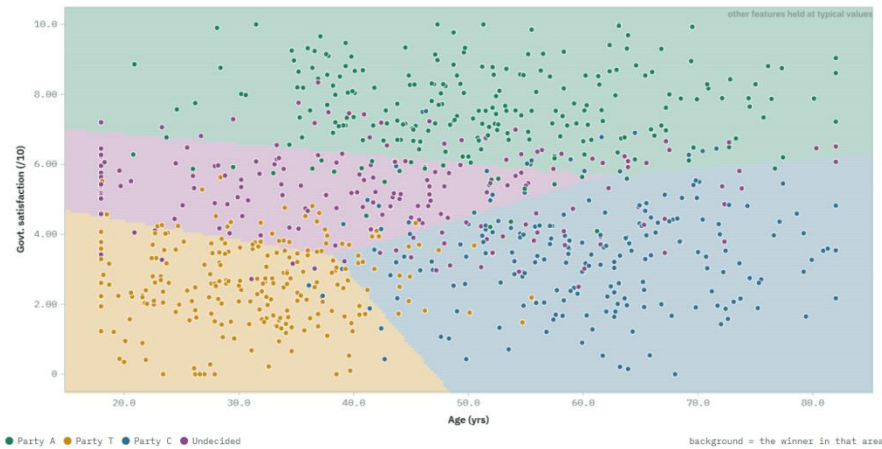
Undecided

1.28 pts = base points
 -0.027×28.0 (Age) = -0.767
 $+0.359 \times 0.800$ (Govt. satisfaction) = $+0.287$
 $-0.185 \times \text{Yes (Urban voter)}$ = -0.185
 = **0.615 pts**

highest score wins → **Party T**

ALGORITHM Linear Regression Classification DATASET Election Preference (Multiclass)

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Step 2a – convert each score to e^{score} (always positive)

Party A e^{-9.92} = 0.000
 Party T e^{7.92} = 2741
 Party C e^{1.62} = 5.05
 Undecided e^{0.615} = 1.85

Total (add all 4 up) = 2748

Step 2b – each class's share = its own e^{score} ÷ that same total (2748)

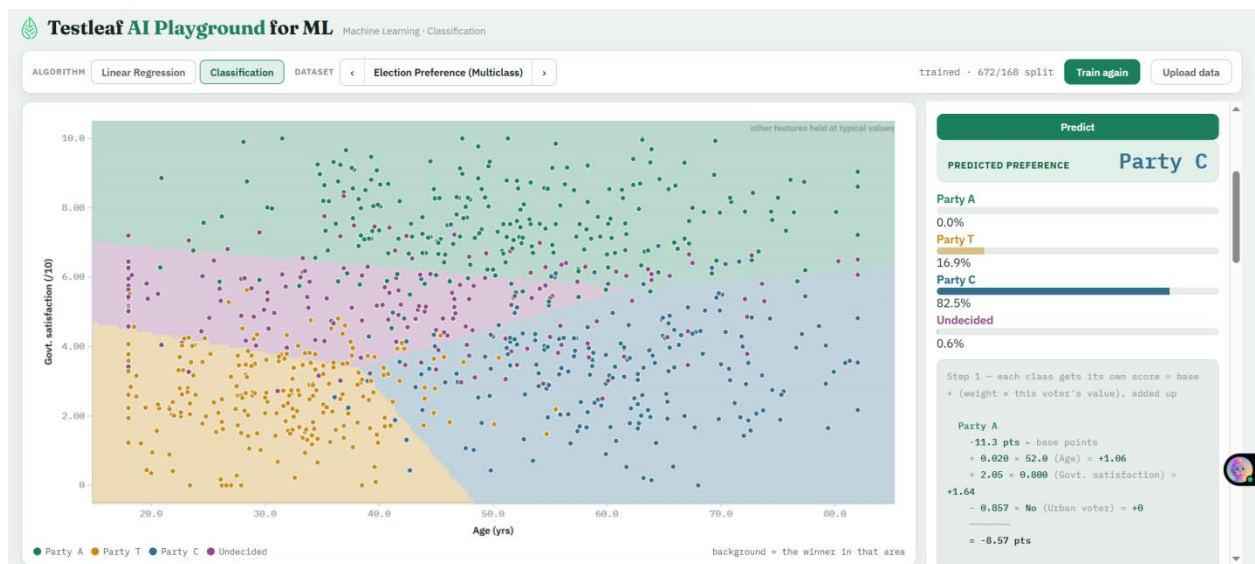
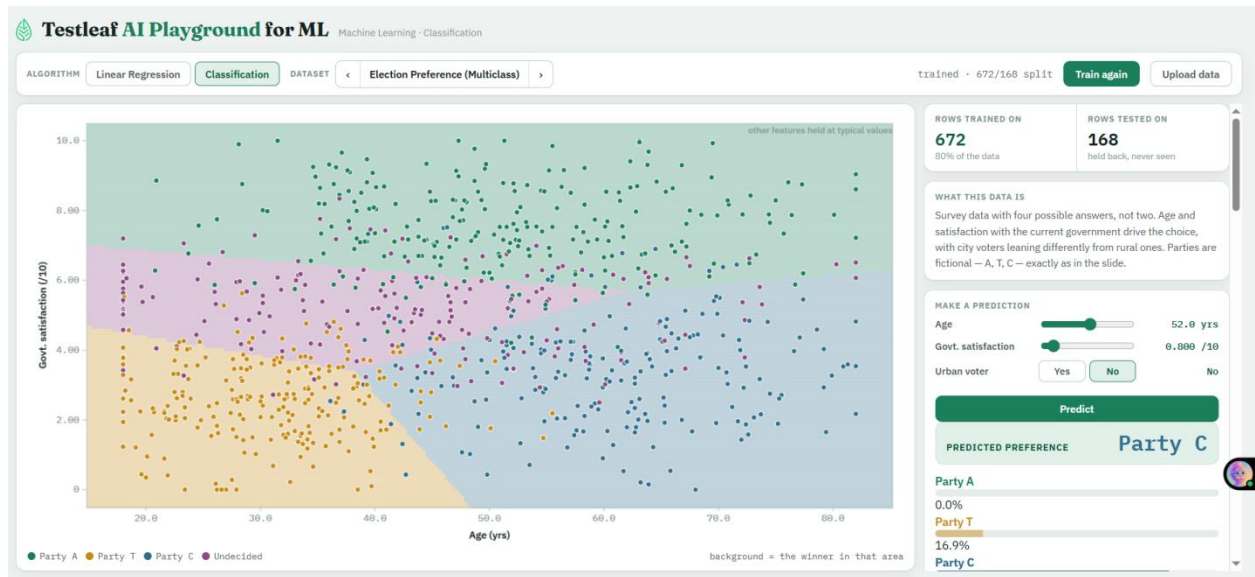
Party A 0.000 ÷ 2748 = 0.00
 Party T 2741 ÷ 2748 = 99.7%
 Party C 5.05 ÷ 2748 = 0.2%
 Undecided 1.85 ÷ 2748 = 0.1%

All 4 shares add back up to 100% – that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

THE TRAINED MODEL

Party A score = -11.3 pts = base points

b) Age >50 and Govt satisfaction less ,non urban



ALGORITHM Linear Regression Classification DATASET Election Preference (Multiclass)

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other features held at typical values

= -8.57 pts

Party T

11.5 pts + base points

- 0.130 × 52.0 (Age) = -6.78

- 1.49 × 0.800 (Govt. satisfaction) = -1.19

+ 1.30 × No (Urban voter) = +0

= 3.49 pts

Party C

0.764 pts + base points

+ 0.897 × 52.0 (Age) = +5.04

- 0.902 × 0.800 (Govt. satisfaction) = -0.721

- 1.14 × No (Urban voter) = +0

= 5.08 pts

Undecided

1.28 pts + base points

- 0.027 × 52.0 (Age) = -1.42

+ 0.359 × 0.800 (Govt. satisfaction) = +0.287

- 0.185 × No (Urban voter) = +0

ALGORITHM Linear Regression Classification DATASET Election Preference (Multiclass)

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other features held at typical values

+ 0.359 × 0.800 (Govt. satisfaction) = +0.287

- 0.185 × No (Urban voter) = +0

= 0.143 pts

highest score wins - Party C

Step 2a - convert each score to e^{score} (always positive)

Party A e^{-8.57} = 0.000

Party T e^{3.49} = 32.8

Party C e^{5.08} = 160

Undecided e^{0.143} = 1.15

Total (add all 4 up) = 194

Step 2b - each class's share = its own e^{score} ÷ that same total (194)

Party A 0.000 ÷ 194 = 0.0%

Party T 32.8 ÷ 194 = 16.9%

Party C 160 ÷ 194 = 82.5%

Undecided 1.15 ÷ 194 = 0.6%

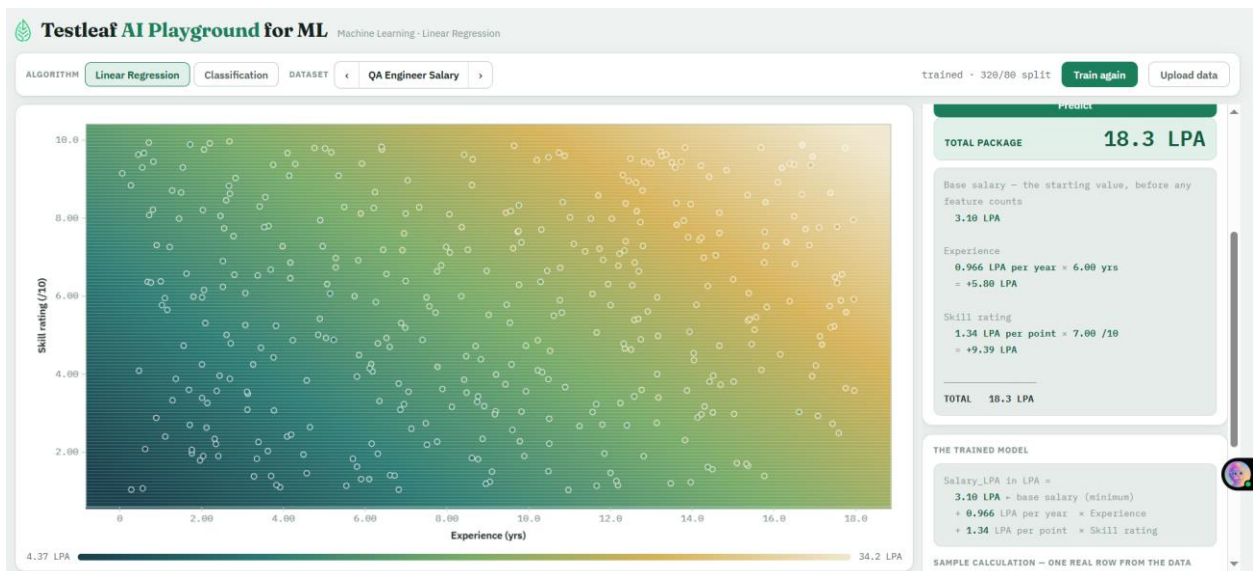
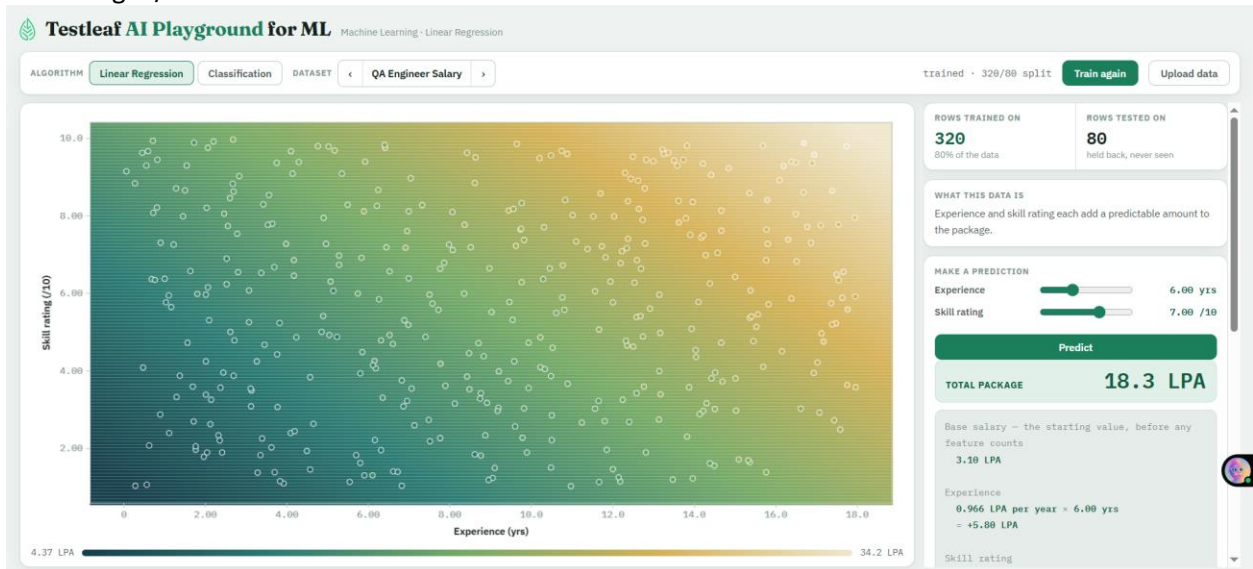
All 4 shares add back up to 100% - that's the whole point of dividing by the same total. e

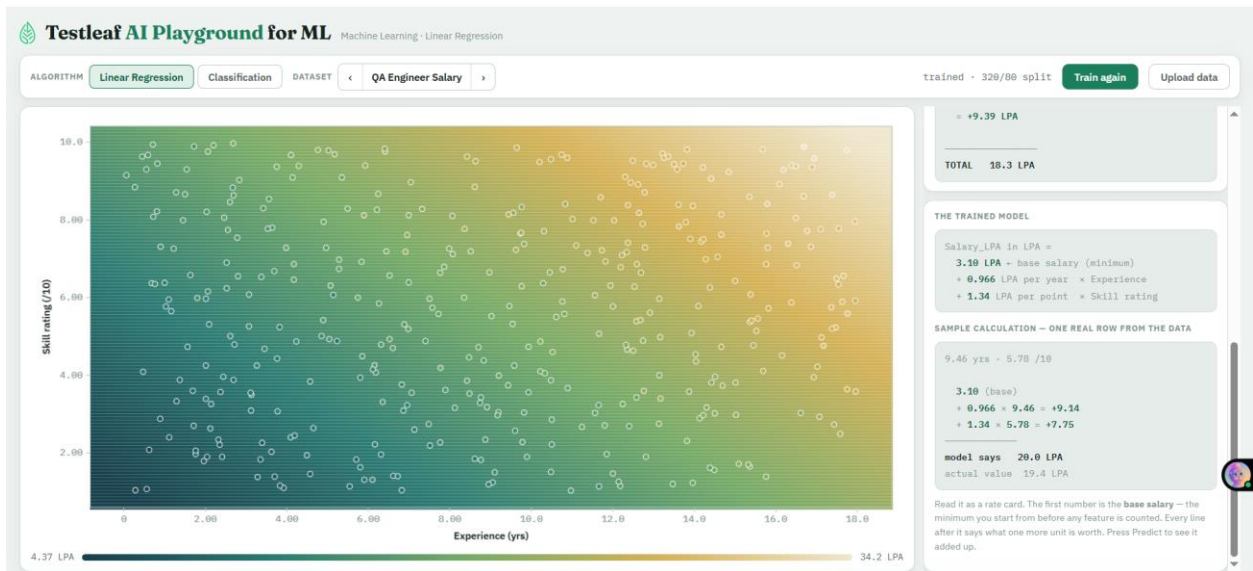
2) Linear regression

a) QA Salary

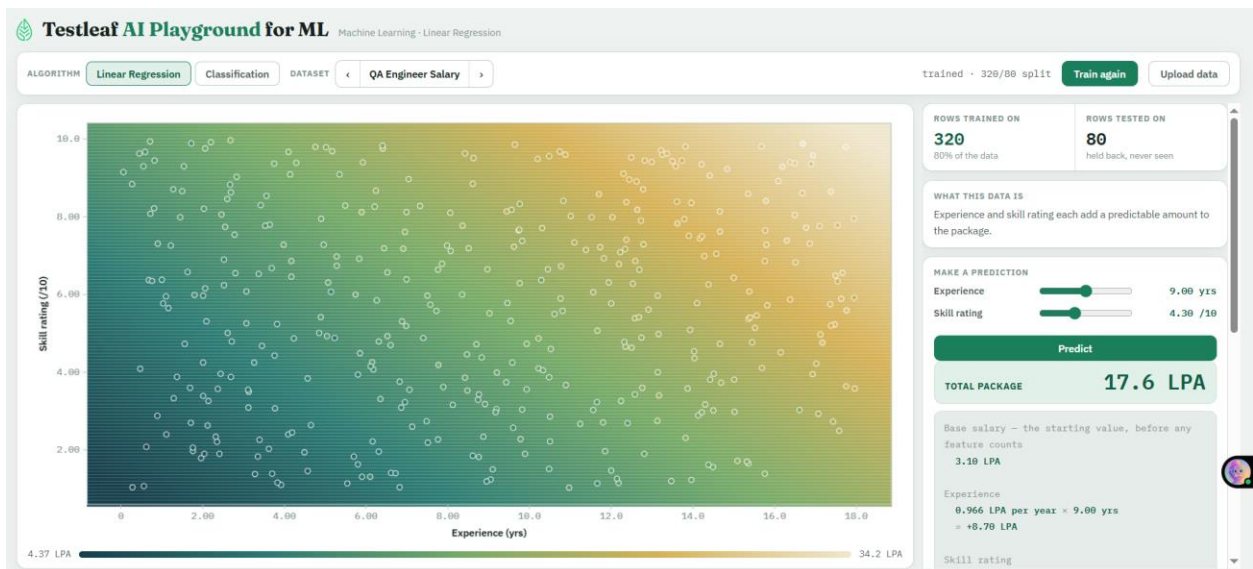
Experience: 6 years

Skill rating: 7/10



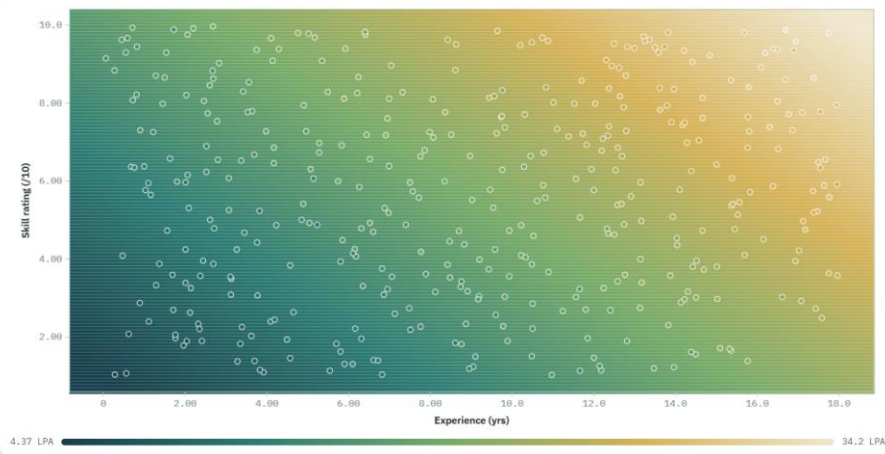


Experience: 9 years
Skill rating: 4.30/10



ALGORITHM **Linear Regression** Classification DATASET < QA Engineer Salary >

trained · 320/80 split **Train again** Upload data



Predict

TOTAL PACKAGE 17.6 LPA

Base salary – the starting value, before any feature counts
3.10 LPA

Experience
0.966 LPA per year × 9.00 yrs
= +8.70 LPA

Skill rating
1.34 LPA per point × 4.30 /10
= +5.77 LPA

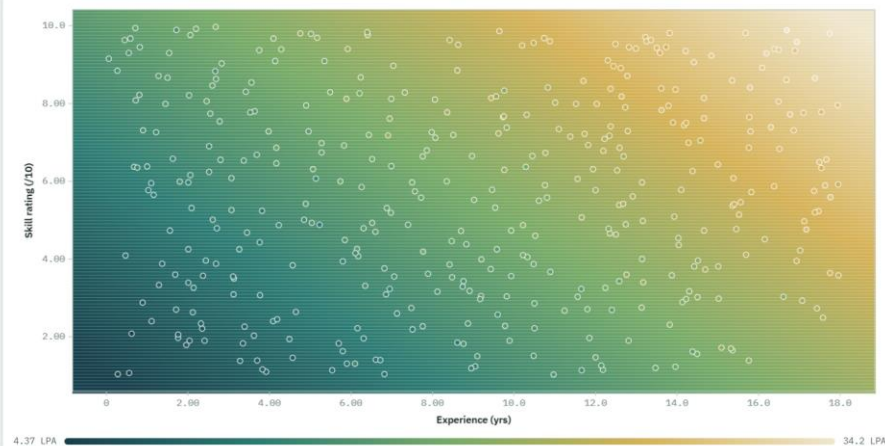
TOTAL 17.6 LPA

THE TRAINED MODEL

Salary_LPA in LPA =
3.10 LPA ← base salary (minimum)
+ 0.966 LPA per year × Experience
+ 1.34 LPA per point × Skill rating

ALGORITHM **Linear Regression** Classification DATASET < QA Engineer Salary >

trained · 320/80 split **Train again** Upload data



= +5.77 LPA

TOTAL 17.6 LPA

THE TRAINED MODEL

Salary_LPA in LPA =
3.10 LPA ← base salary (minimum)
+ 0.966 LPA per year × Experience
+ 1.34 LPA per point × Skill rating

SAMPLE CALCULATION – ONE REAL ROW FROM THE DATA

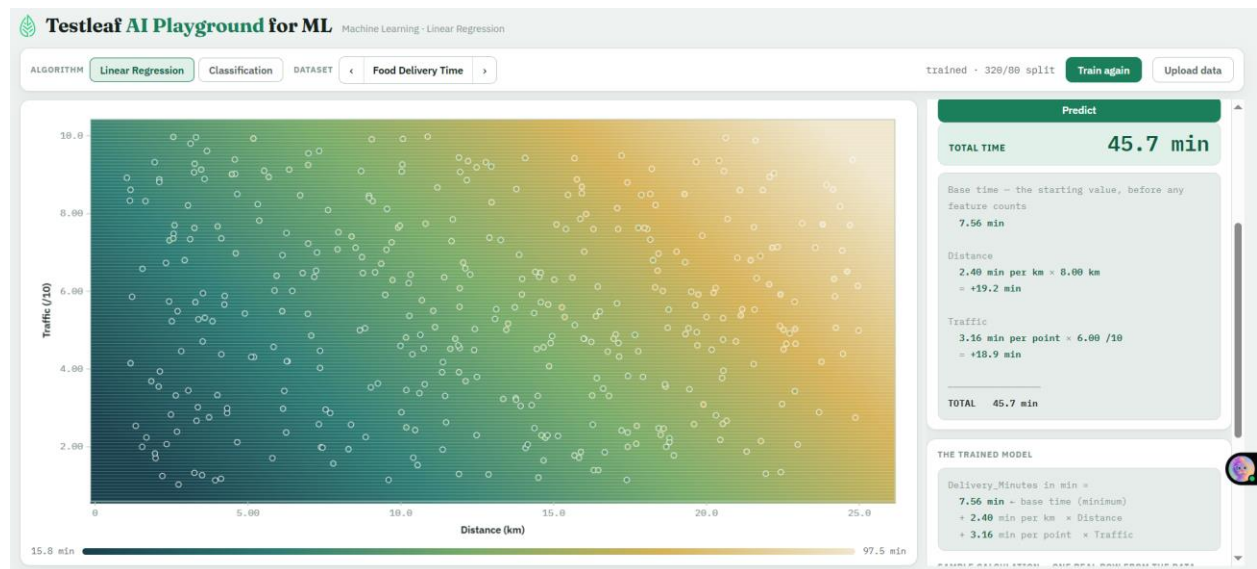
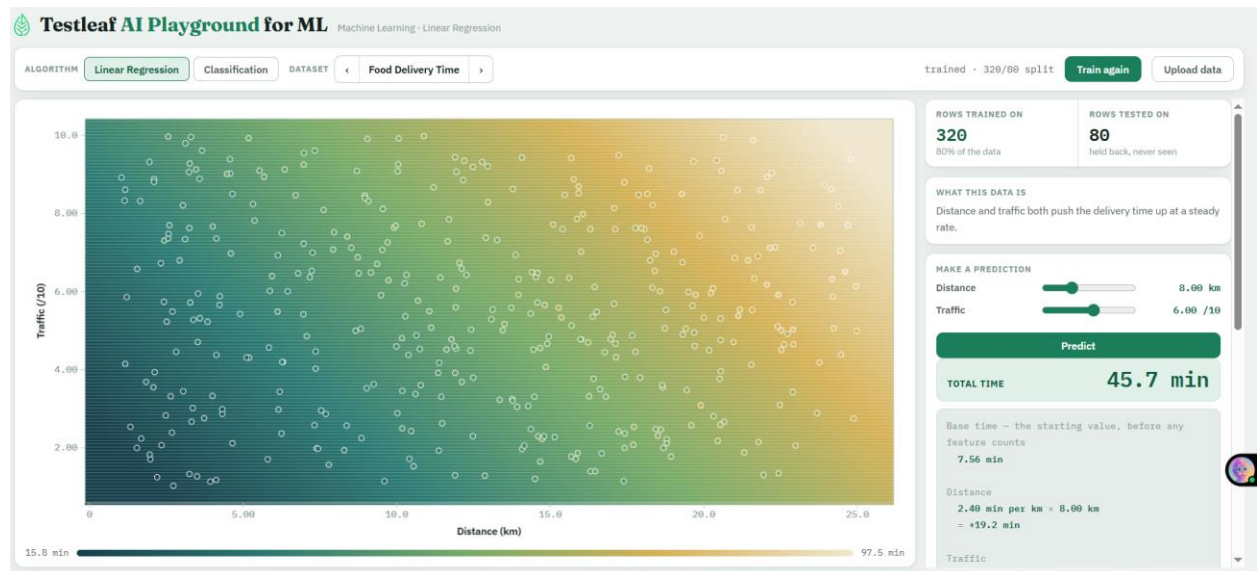
9.46 yrs · 5.78 /10

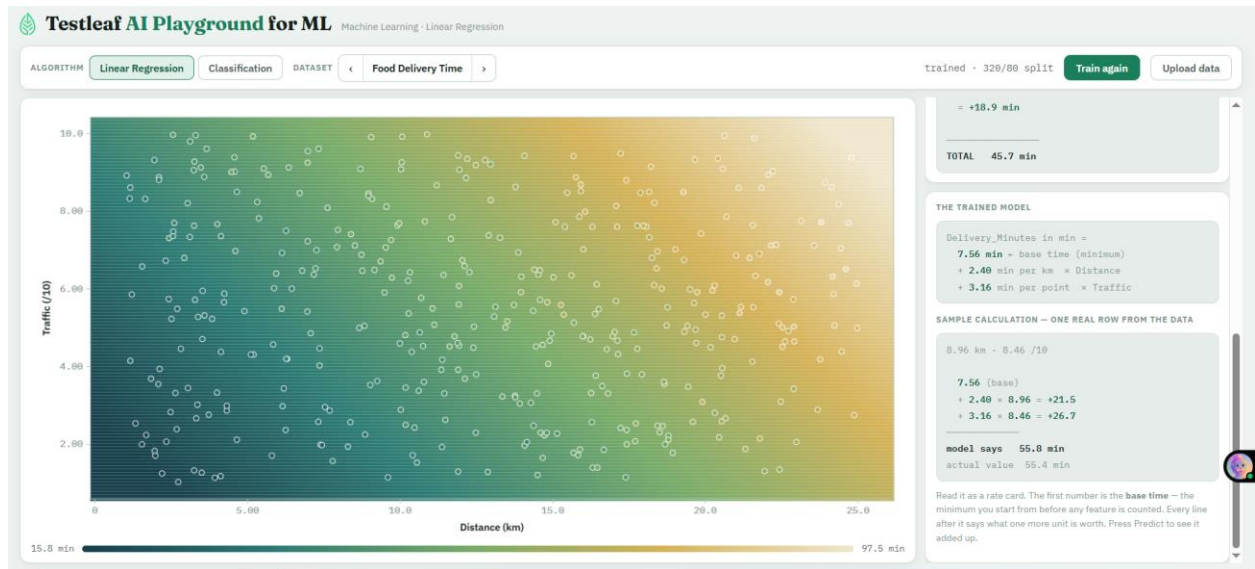
3.10 (base)
+ 0.966 × 9.46 = +9.14
+ 1.34 × 5.78 = +7.75

model says 20.0 LPA
actual value 19.4 LPA

Read it as a rate card. The first number is the **base salary** – the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.

Distance: 8 KM
Traffic: 6/10





Distance: 17.5 KM
Traffic: 4.2/10

